

5 A Spread-Spectrum Synchronous Buck Module with a Two-Stage Output Filter: Design and Experimental Characterization

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Abstract—This paper presents the design and experimental characterization of a 5 A step-down (buck) power module intended as a higher-performance, pin- and form-compatible replacement for the ubiquitous LM2596 module. The converter is built around the Diodes AP64501 synchronous buck (3.8–40 V input, peak-current-mode control at 570 kHz, integrated 45 mΩ/20 mΩ power MOSFETs and ± 6 % frequency spread spectrum) and is implemented on a four-layer printed circuit board with a continuous return-reference plane and a thermal via array beneath the exposed pad. A distinctive feature is a two-stage output filter in which the control-loop feedback is deliberately sampled *before* the second-stage LC network, preserving the manufacturer reference type-II compensation while relegating the post-filter to a passive role. The module was characterized at $V_{in} \approx 12$ V, $V_{out} \approx 4.8$ V, up to 1.155 A. Measured peak efficiency is 87.8 %, output ripple is 7.8 mV_{pp} (≈ 0.16 % of V_{out}), the hotspot temperature is 34.8 °C ($\Delta T \approx 11$ °C), and radiated emissions remain below the displayed CISPR 32 envelope with ≈ 3 dB worst-case margin near 200 MHz. The 2.23 % load regulation is shown—analytically and by inspection of the schematic—to originate almost entirely from the DC resistance of the out-of-loop second-stage inductor rather than from the regulation loop. Design trade-offs and recommended follow-up measurements are discussed.

Index Terms—Synchronous buck converter, AP64501, spread-spectrum modulation, electromagnetic interference (EMI), output-voltage ripple, two-stage LC filter, load regulation, output impedance, power-distribution network, CISPR 32, four-layer PCB.

Manuscript compiled June 14, 2026 (Rev. C). The author is with Electgpl, Buenos Aires, Argentina (<https://github.com/electgpl>). Design performed in KiCad 9.0.4; device under test based on the Diodes AP64501SP-13. Prototype labeled “For evaluation only — not FCC approved for resale.” Ambient temperature for the thermal measurement was not logged and is assumed ≈ 24 °C where noted.

I. Introduction

The LM2596-based step-down module is one of the most widely deployed power bricks in hobbyist and low-volume industrial designs, owing to its low cost and convenient form factor. Its limitations are equally well known: an asynchronous topology with a Schottky catch diode, a switching frequency near 150 kHz, modest light-load efficiency, comparatively high output ripple, and no provision for electromagnetic-interference (EMI) mitigation. This work targets a mechanically compatible replacement that retains the familiar IN+/GND/OUT/EN interface while substantially improving switching performance, output ripple, thermal behavior and radiated emissions.

The design is built around the Diodes AP64501, a 5 A synchronous buck converter with a 3.8–40 V input range, peak-current-mode control at a fixed 570 kHz internal clock, fully integrated 45 mΩ (high-side) and 20 mΩ (low-side) power MOSFETs, a proprietary gate-driver scheme that controls switch-node ringing, and frequency spread spectrum (FSS) with ± 6 % jitter for EMI reduction [1]. The contribution of this paper is twofold: (i) a complete, reproducible design record (schematic, bill of materials and four-layer stack-up) of an EMI-aware module, and (ii) a full experimental characterization—efficiency, output ripple, thermal imaging and radiated emissions—together with an analysis of the two-stage output-filter trade-off that explains the measured efficiency and load-regulation figures.

II. Power-Stage Architecture and Schematic

A. Integrated Converter

The AP64501 employs fixed-frequency peak-current-mode control with an internal $0.8\text{ V} \pm 1\%$ reference, programmable soft-start, cycle-by-cycle current limit with entry into hiccup mode after 512 consecutive limit events, output over-voltage protection, EN-programmable under-voltage lockout and thermal shutdown [1]. The device is supplied in an SO-8EP package whose exposed pad provides the primary thermal path.

B. Two-Stage Power Train and Capacitance Partition

The converter migrates the LM2596 interface from a 150 kHz asynchronous stage to a 570 kHz synchronous one and adds a second-stage output filter. It is essential to distinguish which output capacitance is inside and outside the control loop, since this governs both stability and load regulation. Table I summarizes the partition.

TABLE I
Output Capacitance Partition and Loop Relationship

Block	Components	Relationship to control loop
Input	$C1, C2 = 2 \times 10\text{ }\mu\text{F} + C3\text{ }100\text{ nF}$	Input decoupling / input switching loop
First stage (buck)	$L1\text{ }3.6\text{ }\mu\text{H} + C8\text{ }100\text{ nF} + C9/C10/C11 = \mathbf{66\text{ }\mu\text{F}}$	Regulated node — feedback is sensed here
Second stage (post-filter)	$L2\text{ }10\text{ }\mu\text{H} + C12\text{ }22\text{ }\mu\text{F}$	Outside the loop — passive ripple/EMI attenuation

The first-stage node presents $66\text{ }\mu\text{F}$ of ceramic capacitance—identical to the $3 \times 22\text{ }\mu\text{F}$ C_{out} of the manufacturer reference design [1]—so the reference type-II compensation ($R2 = 15.8\text{ k}\Omega$, $C6 = 2.7\text{ nF}$) remains valid. The post-filter ($L2, C12$) is added downstream of the feedback tap and therefore does not alter the loop, at the cost of the DC drop of $L2$ (Section V-C).

C. Feedback Sensed Before the Filter

A design note on the schematic states explicitly that the feedback sample is taken before the LC filter, so as not to alter the compensation phase of the converter nor add a new pole. This is a deliberate and well-founded choice: it keeps the loop identical to the reference (stable, with $66\text{ }\mu\text{F}$ and type-II compensation) and leaves the second stage as a purely passive filter. The output is made adjustable by the divider

$$V_{\text{out}} = V_{\text{ref}} \cdot (1 + R_{\text{top}} / R_{\text{bot}}) \quad (1)$$

with upper branch $R4$ ($100\text{ }\Omega$) in series with trimmer $RV1$ ($47\text{ k}\Omega$) taken from the regulated node, lower resistor $R3$ ($2.2\text{ k}\Omega$) to ground, and a feed-forward capacitor $C7$ (10 nF) across the upper branch to introduce a phase-lead zero. With $V_{\text{ref}} = 0.8\text{ V}$ and $R3 = 2.2\text{ k}\Omega$, the measured 4.86 V set-point corresponds to $R_{\text{top}} \approx 11\text{ k}\Omega$ ($RV1$ at roughly one quarter of travel); the theoretical adjustment range is $\approx 0.84\text{ V}$ to $\approx 18\text{ V}$, bounded in practice by V_{in} headroom.

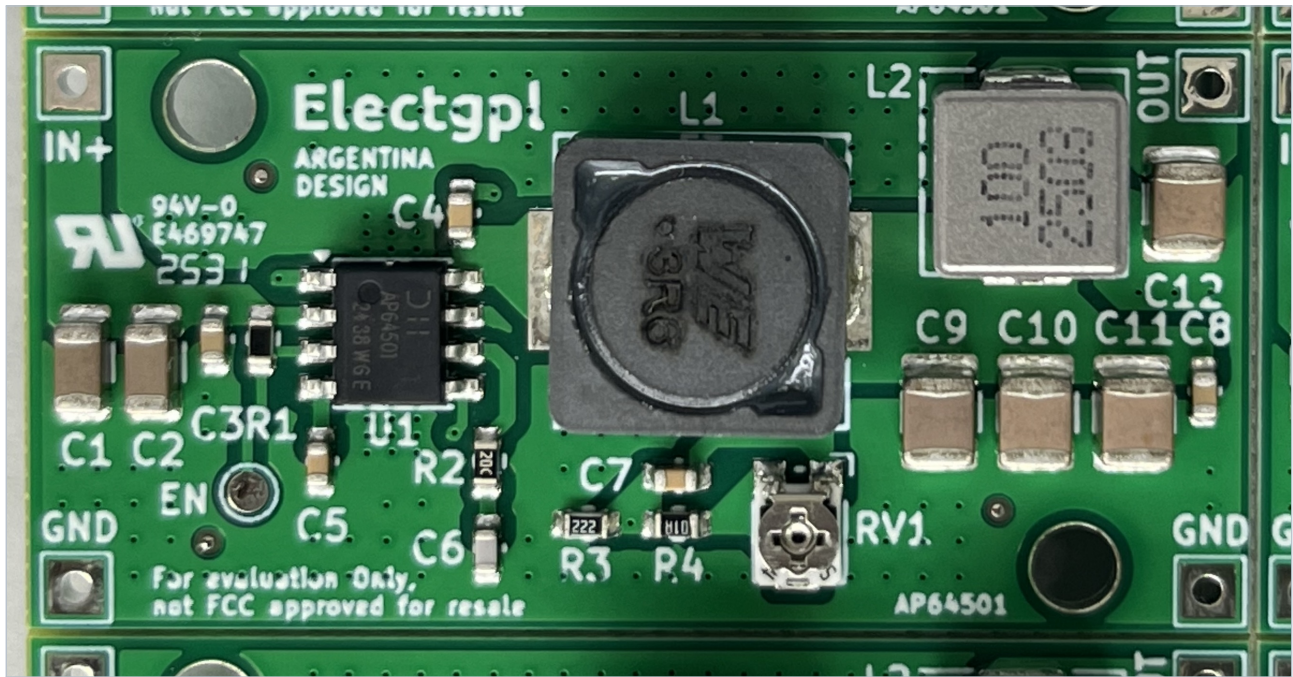


Fig. 1. Assembled module. U1 = AP64501 (SO-8EP); L1 = WE 3R6 (3.6 μ H) main inductor; L2 = 10 μ H post-filter inductor; C9-C11 form the 66 μ F first-stage bank (loop node); C12 is the second-stage capacitor; RV1 sets V_{out} .

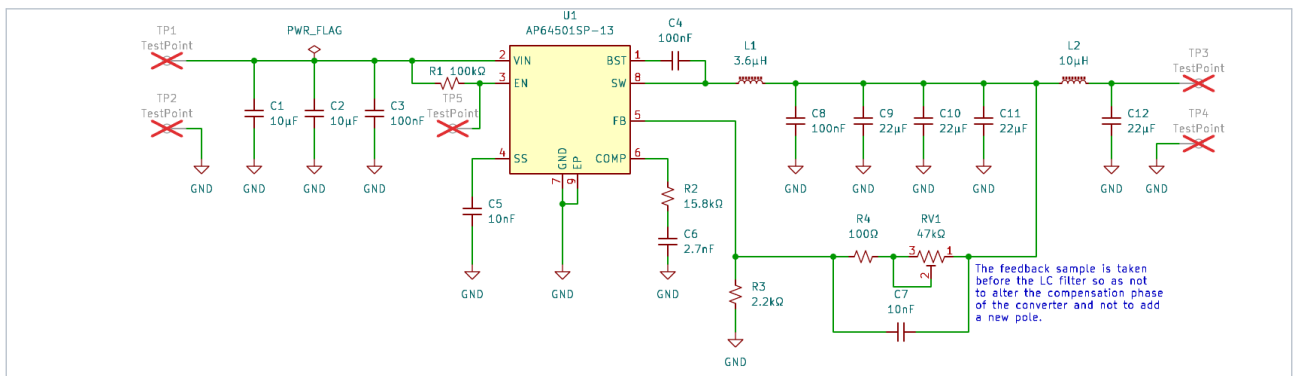


Fig. 2. Schematic (KiCad 9.0.4, Rev 1.0). The chain L1 $\rightarrow$ (loop node, FB) $\rightarrow$ L2 $\rightarrow$ C12 $\rightarrow$ OUT and the design note fixing the feedback tap ahead of the LC filter are highlighted by the topology.

TABLE II
Bill of Materials

Ref.	Value	Function
U1	AP64501SP-13	5 A synchronous buck (SO-8EP, EP to GND)
L1	3.6 μ H (WE 3R6)	Main inductor — first stage
L2	10 μ H	Post-filter inductor — second stage
C1, C2	10 μ F each	Input capacitors (ceramic)
C3	100 nF	Input high-frequency decoupling
C4	100 nF	Bootstrap (BST-SW)
C5	10 nF	Soft-start (SS)
C6	2.7 nF	Type-II compensation (COMP)
C7	10 nF	Feedback feed-forward
C8	100 nF	First-stage output, high frequency
C9, C10, C11	22 μ F each (66 μ F)	First-stage bulk (loop node)
C12	22 μ F	Second-stage output (post-filter, out of loop)
R1	100 k Ω	EN pull-up / UVLO setting
R2	15.8 k Ω	Type-II compensation (COMP)
R3	2.2 k Ω	Feedback divider (lower)
R4	100 Ω	Feedback divider (upper series)
RV1	47 k Ω (trimmer)	Output-voltage adjustment
TP1-TP5	—	Test points (V_{in} , EN, V_{out} nodes)
FID1-FID3	—	Assembly fiducials

III. Printed-Circuit Implementation

The board is realized on four copper layers, a decision central to the measured power-integrity and EMI performance. Table III lists the function of each layer as inferred from the copper artwork (Fig. 3).

TABLE III
Function of Each Copper Layer

Layer	Design observation	Implication
TOP	Components and power copper: V_{in} input, a compact SW node, the first-stage (loop) node and OUT.	Input switching loop and SW node of minimal area $\rightarrow$ reduced di/dt radiating area.
INNER1	Continuous reference (GND) plane directly beneath TOP.	Low-inductance image-current return for every power trace $\rightarrow$ key to the clean CISPR scan.
INNER2	Additional plane (GND/power) with stitching-via field.	Reinforces the PDN and inter-layer shielding; low plane impedance.
BOTTOM	Sensitive signal routing (feedback) plus GND fill.	Feedback routed away from SW, referenced to an inner plane $\rightarrow$ low noise on the regulated node.

Two cross-cutting features stand out: a dense GND stitching-via field that ties the planes together across all layers, providing return continuity and shielding; and a thermal via array beneath the exposed pad of U1 (white outline in Fig. 3), which channels heat from the EP into the inner and bottom copper—consistent with the ≈ 12 $^{\circ}\text{C}/\text{W}$ effective thermal resistance estimated in Section V-E. The continuous return plane on INNER1 under the TOP power copper, the compact SW node and the device FSS jointly account for the low radiated-emission levels

(Section V-F).

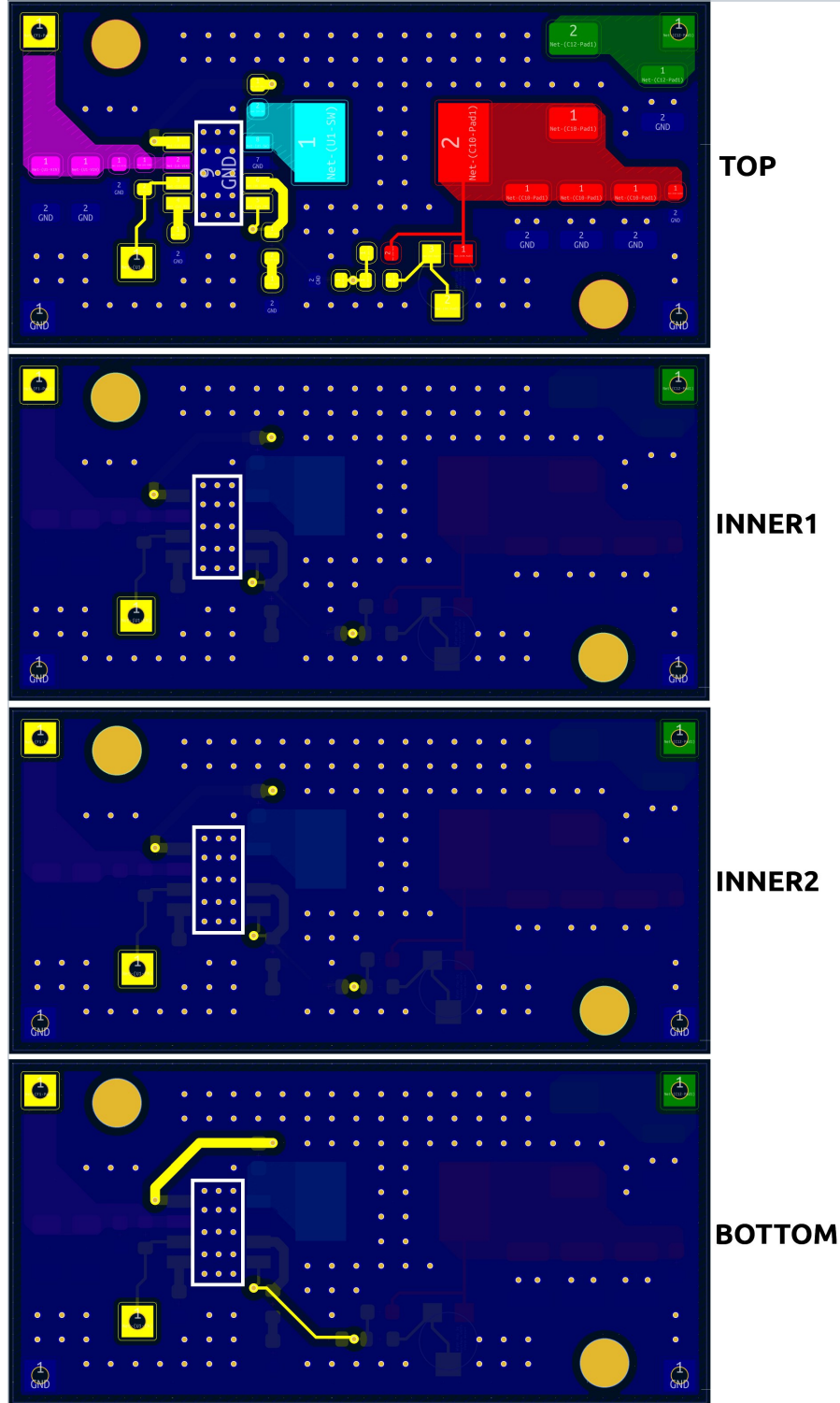


Fig. 3. Four-layer artwork (TOP / INNER1 / INNER2 / BOTTOM). White outline: thermal/GND via array beneath the U1 exposed pad. Yellow dots: the GND stitching-via field.

IV. Experimental Method

The module was supplied from a bench source at $V_{in} \approx 12$ V (logged average 11.879 V) with the output trimmed to ≈ 4.8 V. An electronic load swept the output current from 0 to 1.155 A in 30 steps, averaging 16 readings per point with a 0.25 s settling interval. Table IV lists the instrumentation and conditions.

TABLE IV
Test Conditions and Instrumentation

Quantity / instrument	Configuration
Input voltage	$V_{in} \approx 12$ V (measured 11.879 V), bench supply
Output voltage	≈ 4.8 – 4.86 V, set by RV1
Load sweep	Electronic load, 30 steps, 0–1.155 A, 0.25 s settle, 16-sample average
Bench power budget	$P_{budget} = 15$ W (caps the swept I_{out})
Ripple	Rigol oscilloscope, CH1 5 mV/div, statistical <i>Ripple</i> function (1000 counts), 20 MHz bandwidth limit
Thermal	IR camera (iron palette), board in still air, no heatsink or forced flow
EMC	Radiated emissions 30 MHz–1 GHz against a CISPR 32 envelope (40/47 dB μ V/m)

The sweep terminated at 1.155 A because of the 15 W bench budget, not a device limitation: the automated report records “3 %/5 %/10 % drop: not reached” and “no cutoff,” meaning the output never sagged enough to trigger the drop thresholds and the AP64501 current limit was not reached. The present campaign therefore covers operation up to ≈ 1.16 A; determining the true maximum output current (toward the 5 A device rating) requires a higher-power bench and is left as future work (Section VI).

V. Results and Discussion

Table V collects the headline measured figures; the subsections that follow analyze each result.

TABLE V
Summary of Measured Performance

Parameter	Result	Condition / note
V_{out} , no load	4.861 V	$I_{out} = 0$ (trimmed by RV1)
V_{out} , full load	4.752 V	$I_{out} = 1.155$ A
Load regulation	2.23 % ($R_{out,eq} \approx 94$ m Ω)	0–1.155 A; root cause in V-C
Max. swept I_{out}	1.155 A	bench-limited; I_{lim} not reached
Peak / full-load efficiency	87.8 % @0.84 A / 87.0 % @1.16 A	max. loss 0.90 W
Output ripple (pp, BW 20 MHz)	7.76 mV typ. / 11.84 mV max	≈ 0.16 % of V_{out}
Thermal hotspot	34.8 °C	$\Delta T \approx 11$ °C over assumed ~ 24 °C ambient
Radiated emissions	Below 40/47 dB μ V/m envelope	≈ 3 dB worst-case margin (~ 200 MHz)

A. Efficiency and Power Balance

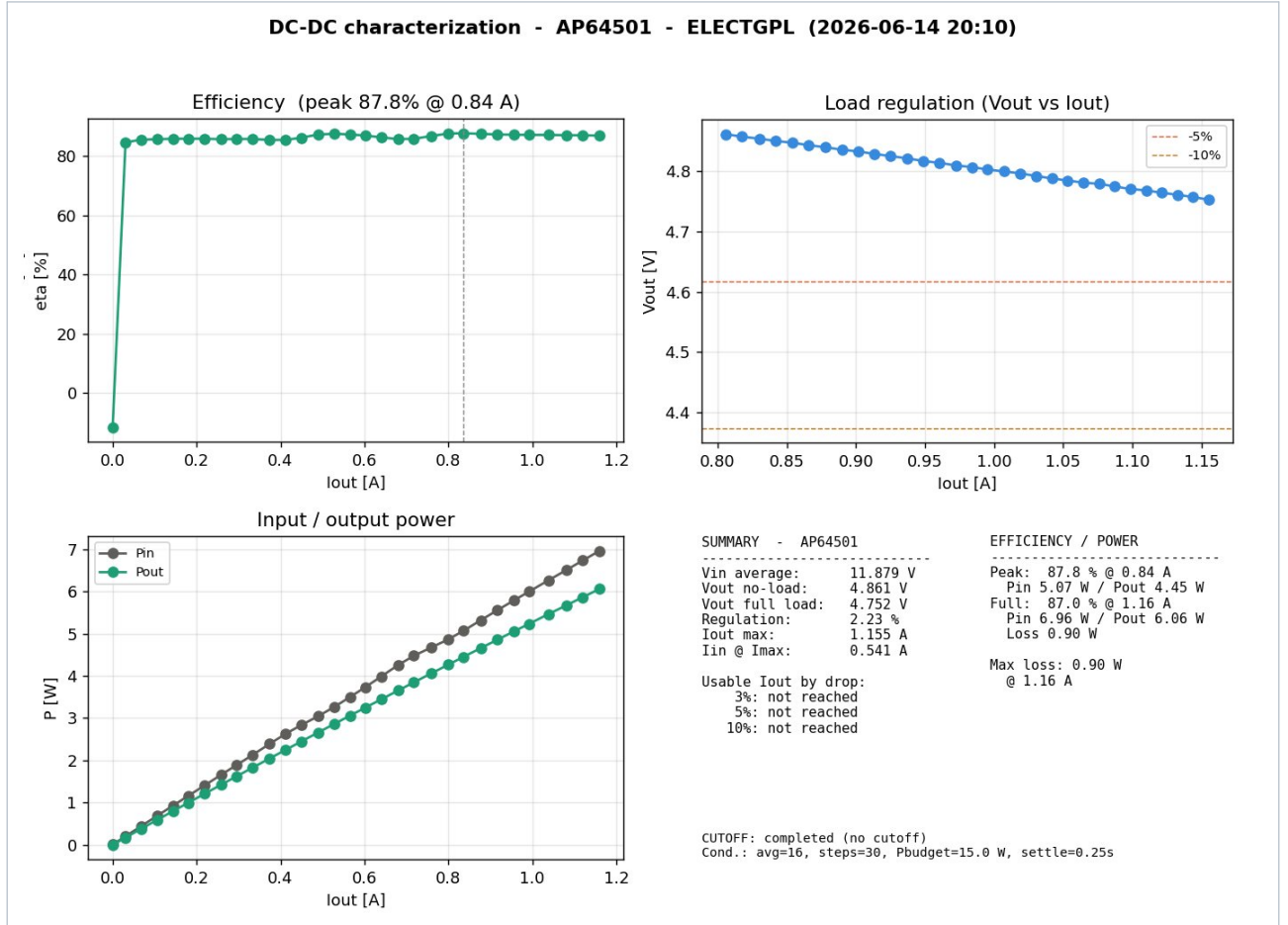


Fig. 4. Automated DC-DC characterization: efficiency $\eta(I_{out})$, load regulation $V_{out}(I_{out})$ and input/output power, with the bench numeric summary at right.

Efficiency is flat at 86–88 % across the continuous-conduction region, peaking at 87.8 % at 0.84 A and reaching 87.0 % at full load (1.16 A), where the total loss is 0.90 W. The first sweep point ($I_{out} \rightarrow 0$) falls below zero by definition of $\eta = P_{out}/P_{in}$ at no load ($P_{out} = 0$ while the device still draws switching and quiescent current); it is an artifact, not a physical efficiency.

For the 12 V→4.8 V operating point the bare-IC reference design exceeds 90 % [1],[2]; the module delivers ≈ 87 –88 %, a 3–6-point module-level deficit. With the bill of materials now confirmed, the dominant contributors are, in order: (i) the DC resistance of the out-of-loop second-stage inductor L2 (10 μ H), which carries the full DC output current—at DCR ≈ 80 m Ω this is $I^2 \cdot \text{DCR} \approx 0.11$ W at 1.16 A, about 12 % of the measured loss; (ii) the DCR of L1 and interconnect; and (iii) the operating point, since 1.16 A is well below the peak-efficiency load of a 5 A converter, so fixed losses weigh proportionally more. Conduction losses in the integrated MOSFETs are bounded: high-side $\approx D \cdot I^2 \cdot R_{HS} = 0.40 \cdot 1.16^2 \cdot 0.045 \approx 24$ mW and low-side $\approx 0.60 \cdot 1.16^2 \cdot 0.020 \approx 16$ mW. The trade-off is consistent with the design intent: the post-filter trades a few points of efficiency for ripple and EMI.

B. Inductor Ripple, CCM and Filter Corners

The peak-to-peak ripple current of the main inductor in steady state is

$$\Delta I_L = V_{out}(V_{in} - V_{out}) / (V_{in} L_1 f_{sw}) \quad (2)$$

which evaluates to $\Delta I_L \approx 1.40$ A_{pp} (≈ 28 % of the 5 A rating; the usual 20–40 % guideline). The inductor-current valley at 1.16 A is $1.16 - 0.70 = 0.46$ A > 0 , so the converter operates in continuous conduction (CCM); the CCM/PFM boundary lies at ≈ 0.70 A. Each LC corner follows from

$$f_c = 1 / (2\pi \sqrt{LC}) \quad (3)$$

yielding the values in Table VI.

TABLE VI
Distributed-Element and Filter Parameters

Stage / quantity	Expression	Value
First-stage LC (in loop)	$f_{c1} = 1/(2\pi\sqrt{L1 \cdot 66 \mu F})$, $L1 = 3.6 \mu H$	$\approx 10.3 \text{ kHz}$
Second-stage LC (post-filter)	$f_{c2} = 1/(2\pi\sqrt{L2 \cdot C12})$, $L2 = 10 \mu H$, $C12 = 22 \mu F$	$\approx 10.7 \text{ kHz}$
Second-stage attenuation @570 kHz	$(f_{sw}/f_{c2})^2$ (ideal, before C12 ESR zero)	$\approx 69 \text{ dB}$
Second-stage characteristic impedance	$Z_0 = \sqrt{L2/C12}$	$\approx 0.67 \Omega$

The $\approx 60\text{--}70 \text{ dB}$ of passive attenuation provided by the second stage at the switching frequency is the quantitative explanation for the very low measured ripple (Section V-D). A damping caveat applies: with $Z_0 \approx 0.67 \Omega$ and only the L2 DCR ($\sim 80 \text{ m}\Omega$) as damping, the second-stage quality factor is high (≈ 8) and, being outside the loop, may appear as ringing near 10.7 kHz at the OUT terminal under load steps. This should be verified with a load-transient test (Section VI); if present, an RC snubber across C12 damps it without touching the loop.

C. Load Regulation and Output Impedance

The measured load regulation is 2.23 % (4.861→4.752 V from no load to 1.155 A): a 109 mV drop over 1.155 A, equivalent to a static output resistance

$$R_{\text{out,eq}} = \Delta V_{\text{out}} / \Delta I_{\text{out}} \approx \text{DCR}_{L2} + R_{\text{int}} \approx 94 \text{ m}\Omega \quad (4)$$

The schematic confirms the root cause: feedback is sensed at the first-stage node, by explicit design choice, so the loop regulates that node accurately while the DC resistance of L2 (10 μH) lies outside the loop and appears in full as load droop at OUT. The 94 m Ω figure is fully compatible with the DCR of a compact 10 μH molded inductor ($\approx 60\text{--}90 \text{ m}\Omega$) plus trace and connector resistance—not with a loop deficiency, whose intrinsic DC regulation is $<0.5 \%$.

The decision is technically sound: sensing after L2 would bring the second stage inside the loop, adding an LC pole that erodes phase margin—precisely what the design avoids. Three options exist, in increasing order of intervention: (a) accept and specify the 94 m Ω as the module output impedance (simplest and most honest for a datasheet); (b) reduce the L2 DCR (a larger-section, lower-DCR inductor) to lower the droop without touching the loop; and (c), only if the application demands tight regulation at OUT, sense in Kelvin after L2 with a feed-forward/phase-lead network that restores the high-frequency information of the pre-L2 node and preserves margin, at the cost of added compensation complexity. A four-wire measurement of the L2 DCR is recommended first, to confirm it dominates the 94 m Ω .

D. Output Ripple and Noise

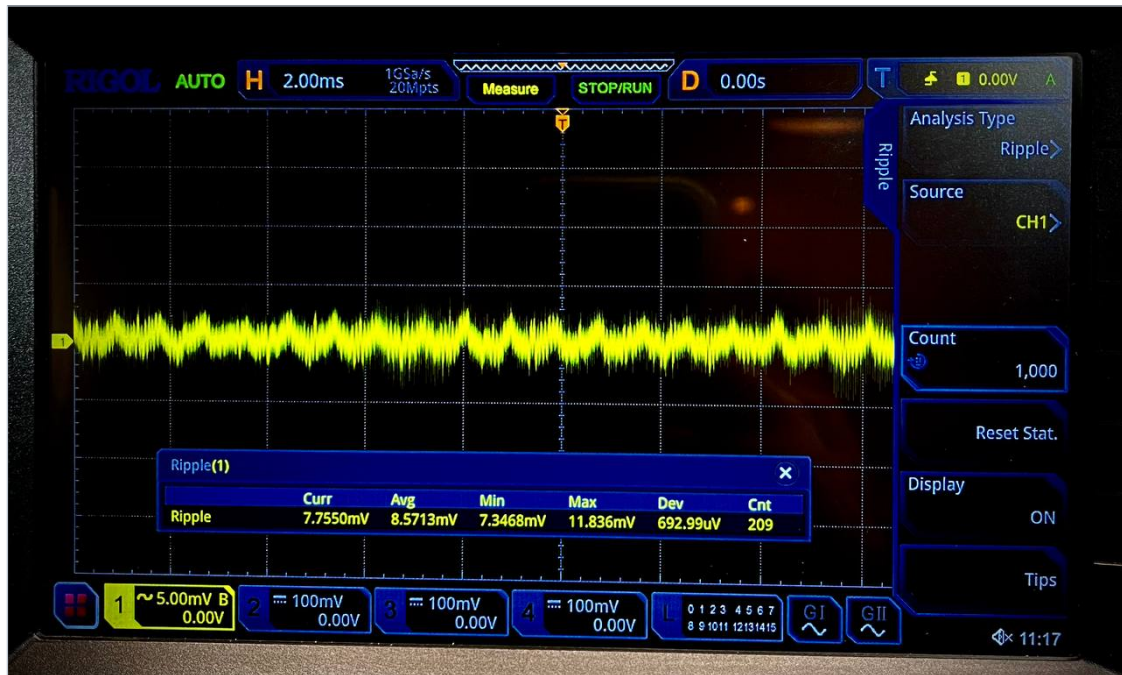


Fig. 5. Output ripple (Rigol, 5 mV/div, 20 MHz bandwidth limit, 1000 counts): V_{pp} 7.755 mV (mean 8.571, max 11.836, σ 0.69 mV).

Output ripple-plus-noise is 7.76 mV_{pp} (mean 8.57, max 11.84), i.e. $\approx 0.16\%$ of V_{out} —a strong result and the direct benefit of the second-stage post-filter (Section V-B: ≈ 69 dB attenuation at f_{sw}). The 20 MHz bandwidth limit (indicator “B”) was engaged, standard practice to bound RF pickup.

Two measurement caveats keep the figure defensible. First, the 2 ms/div timebase does not resolve the 570 kHz fundamental (1.75 μ s period); the instrument’s *Ripple* function computes V_{pp} statistically over the whole record (≈ 20 ms), so the value is valid as peak-to-peak ripple-plus-noise but mixes switching ripple with low-frequency wander and any pickup. Second, a zoomed capture (~ 500 ns/div) with a tip-and-barrel probe across C12—eliminating the alligator ground loop—should confirm the switching-ripple waveform; the ≈ 8 mV_{pp} figure is expected to hold or improve.

E. Thermal Performance

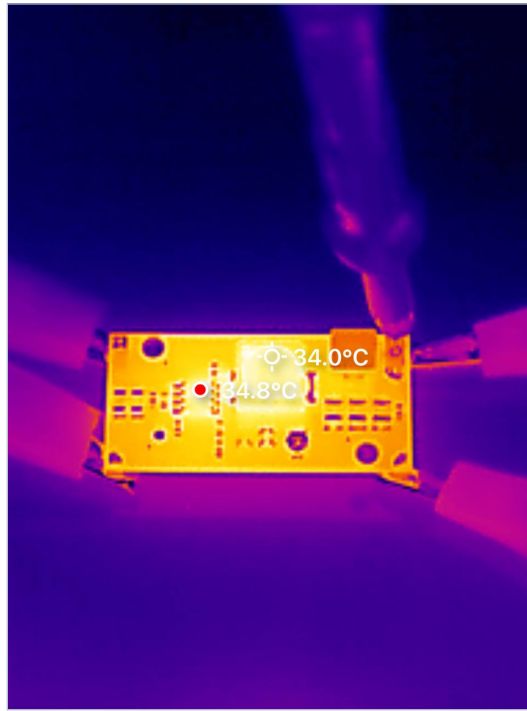


Fig. 6. Thermography. Crosshair 34.0 °C; spot 34.8 °C. Board in still air, no heatsink.

Operating in still air without heatsink or forced ventilation, the hotspot reads 34.8 °C and the board averages ≈ 34.0 °C. Assuming an ambient of ≈ 24 °C (not logged), the rise is $\Delta T \approx 11$ °C for ≈ 0.9 W of loss, giving an effective hotspot-to-ambient thermal resistance of ≈ 12 °C/W—excellent for an SO-8EP with exposed pad on four layers with a thermal via array (Section III). An emissivity caveat applies: IR cameras under-read shiny metal surfaces (inductor can, IC body) owing to low emissivity, whereas readings on the PCB solder mask ($\epsilon \approx 0.9$) are reliable; metallic parts may therefore be somewhat hotter than displayed. Even so, the margin is wide: the temperature confirms ample headroom to operate well above 1.16 A before any thermal limit.

F. Radiated Emissions (CISPR 32)

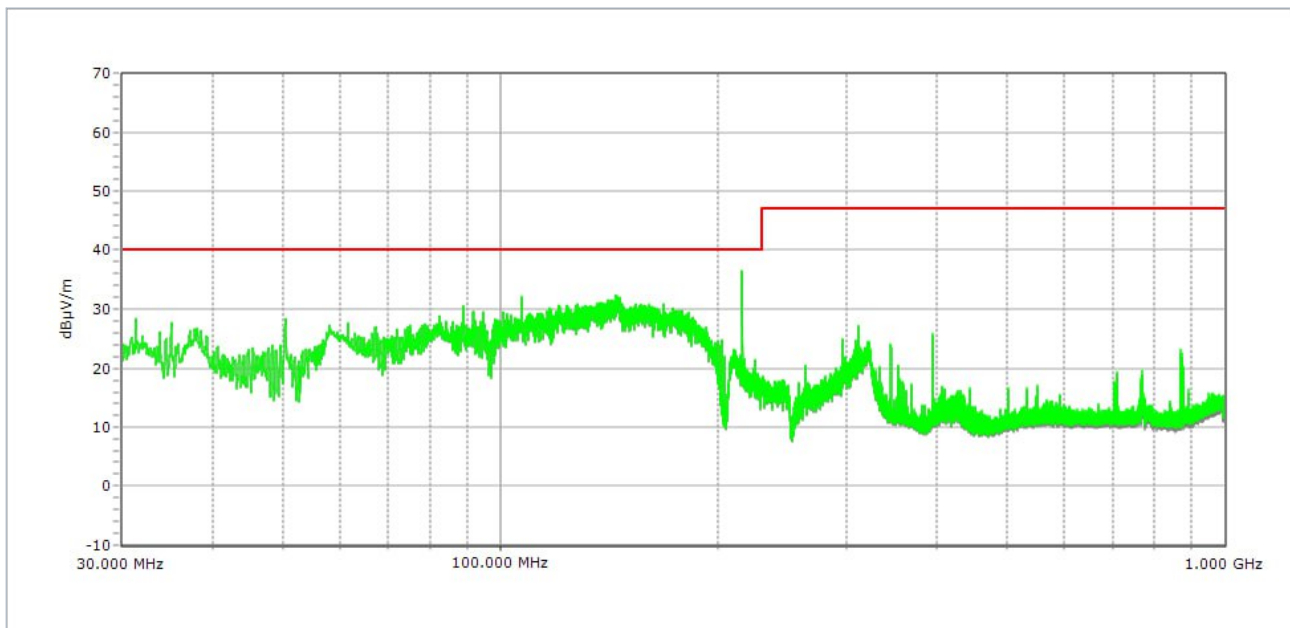


Fig. 7. Radiated emissions 30 MHz–1 GHz (green) against the limit envelope (red): 40 dB μ V/m (30–230 MHz) and 47 dB μ V/m (230 MHz–1 GHz).

The trace remains below the envelope across the band. The displayed limits (40/47 dB μ V/m) correspond to CISPR 32 Class B at 3 m (equivalently Class A at 10 m) [4]. The measured level mostly lies between 20 and 35 dB μ V/m, with a broadband hump between $\approx$ 100 and $\approx$ 200 MHz reaching $\approx$ 37 dB μ V/m—the worst case, at $\approx$ 3 dB margin; above 230 MHz the margin exceeds 10 dB.

This is consistent with the design: the AP64501 FSS (± 6 %) spreads each switching harmonic and lowers peak-detector levels [1]; the continuous return plane on INNER1 under the TOP power copper minimizes di/dt loop area; and the second-stage post-filter attenuates common-mode current toward the output cabling—the principal radiation mechanism in modules of this type. The 100–200 MHz hump has the signature of common-mode current on the I/O cables rather than a direct switching harmonic, and is the feature to watch for additional margin (an input/output ferrite bead or reinforced common-mode filtering). The scan is a pre-compliance peak-detector measurement; the distance, antenna and chamber are not recorded, and quasi-peak/average detectors are not shown. It is adequate to validate the design and estimate margin, but formal conformance requires an accredited measurement.

VI. Conclusion and Recommended Work

A four-layer, EMI-aware 5 A synchronous buck module was designed and characterized as a higher-performance, form-compatible alternative to the LM2596 module. At 12 V $\rightarrow$ 4.8 V it achieves 570 kHz synchronous switching, ≈ 8 mV_{pp} output ripple, a wide thermal margin ($\Delta T \approx 11$ °C for ≈ 0.9 W), and radiated emissions below the displayed CISPR 32 envelope, with no protection event over the swept range. The two seemingly weak figures— ≈ 87 % efficiency and 2.23 % load regulation—are shown to be the quantified, deliberate consequence of a two-stage output filter with pre-filter feedback, not a defect: the load droop is dominated by the out-of-loop L2 DC resistance, and the efficiency deficit is largely the same series resistance plus a light operating point for a 5 A device. Table VII lists the recommended follow-up measurements that would close the remaining open items.

TABLE VII
Recommended Follow-up Measurements

#	Action	Objective
1	Four-wire DCR of L1 and L2; re-test toward I_{lim} /thermal rollback (3–5 A) on a higher-power bench.	Quantify the efficiency/regulation split; establish true maximum current (now bench-limited to 1.16 A).
2	Load-transient (step) test observing OUT and the loop node.	Check for the ≈ 10.7 kHz second-stage ringing ($Q \approx 8$) outside the loop; decide on an RC snubber across C12.
3	Zoomed ripple capture (~ 500 ns/div) with a tip-and-barrel probe across C12.	Quantify true switching ripple, separated from low-frequency content and pickup.
4	If tight regulation is required: lower-DCR L2, or Kelvin sensing after L2 with feed-forward.	Reduce the 94 m Ω output impedance while preserving phase margin.
5	Quasi-peak pre-compliance; mitigate common mode on I/O cabling (ferrite bead) over the 100–200 MHz hump.	Secure margin before an accredited measurement.
6	Thermal sweep vs. load with logged ambient and, where applicable, at $V_{in} = 40$ V.	Map $R\theta$ /derating (a buck's thermal worst case is usually at high V_{in}).

References

- [1] Diodes Incorporated, “AP64501 — 3.8 V to 40 V Input, 5 A Low I_Q Synchronous Buck with Programmable Soft-Start Time,” Datasheet DS41980, Rev. 5. Key parameters: 5 A, peak-current-mode, $f_{sw} = 570$ kHz, $R_{ds(on)} = 45/20$ m Ω , FSS ± 6 %, $V_{ref} = 0.8$ V, reference $C_{out} = 3 \times 22$ μ F, $L = 3.6$ μ H (5 V), SO-8EP.
- [2] Diodes Incorporated, “AP64501SP-EVM — 40 V, 5 A, Low I_Q Synchronous DC-DC Buck Converter Evaluation Board User Guide.”
- [3] Würth Elektronik, AP64501 reference-design data (3.6 μ H “3R6” inductor), we-online.com, ICref / Diodes Incorporated section.

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- [4] CISPR 32:2015 / EN 55032, “Electromagnetic Compatibility of Multimedia Equipment — Emission Requirements,” radiated-emission Class A/B limits (40/47 dB μ V/m, class- and distance-dependent).
 - [5] Good practice for switching-converter ripple measurement: tip-and-barrel / coaxial probing at the output capacitor with a 20 MHz bandwidth limit to bound RF pickup.
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This document was compiled from the KiCad project (schematic Rev 1.0, four-layer artwork) and the bench reports of the device under test (automated DC-DC characterization, thermography, ripple and radiated emissions) dated 2026-06-14. Derived quantities ($R_{\text{out,eq}}$, ΔI_L , LC corners, attenuation, loss balance, effective $R\theta$) were computed from those data under the assumptions stated in each section. The DCR of L1/L2 and the true maximum output current remain to be measured directly.